12

Dry and Transition Cows

METABOLIC AND PHYSIOLOGIC STATUS OF THE TRANSITION COW

Hormonal and Metabolite Changes

The transition period in dairy cows is generally defined as the last 3 weeks of gestation and the first 3 weeks of lactation. The prepartum phase of this period is characterized by rapid fetal growth, mammary gland growth and development, colostrum synthesis, and dramatic changes in endocrine status. Many of the hormonal changes during the péripartum period are to prepare the cow for the substantial increase in energy needs postpartum (Ehrhardt et al., 2016). Plasma concentrations of insulin, insulin-like growth factor 1 (IGF1), and leptin decrease and growth hormone increases as the cow progresses from late gestation to early lactation, with acute changes in plasma concentrations at parturition (Kunz et al., 1985; VandeHaar et al., 1999; Block et al., 2001; Doepel et al., 2002; Rhoads et al., 2004). Insulin-sensitive tissues in peri parturient cows display insulin resistance so that glucose can be directed to the developing fetus and to the mammary gland. Plasma thyroid hormone (T4 and T3) concentrations gradually increase during late gestation, decrease approximately 50 percent at calving, and then begin to increase (Kunz et al., 1985; Pethes et al., 1985). These hormonal responses are designed to increase energy mobilization, reduce basal metabolic rate, and partition nutrients to the mammary gland.

The changes in endocrine status and the decline in dry matter intake (DMI) that usually occurs around parturition influence metabolism and lead to mobilization of fat from adipose tissue and glycogen from the liver. In healthy dairy cows, from about 2 weeks prepartum until 3 days prepartum, plasma fatty acids (FAs) (commonly referred to as nonesterified FAs or NEFAs) increase from <0.2 mEq/L to about 0.3 mEq/L, and then 1 or 2 days before calving, concentrations increase abruptly and usually reach their peak by the second or third day of lactation (often between 0.8 and 1.0 mEq/L). Concentrations then decrease slowly over the next 2 to 3 weeks (Doepel et al., 2002; LeBlanc et al., 2005; Garverick et al., 2013). How much of the initial increase in plasma FAs can be accounted for by changing endocrine status compared with energy restriction resulting from decreased DMI is not known. Force feeding cows during the prefresh period reduced the magnitude of NEFA increase but did not eliminate it (Bertics et al., 1992), meaning that at least part of the prepartum increase is hormonally induced. The rapid rise in NEFA the day of calving is presumably due to stress, but hormonal changes (e.g., elevated growth hormone) could also be involved. In healthy cows, plasma NEFA concentrations decrease rapidly after calving but remain higher than they were before calving for several weeks. In cows that develop health disorders such as a displaced abomasum, prepartum plasma NEFA concentrations often exceed 0.5 mEq/L and can exceed 1.0 mEq/L during the immediate postpartum period (LeBlanc et al., 2005). High rates of lipolysis and the subsequent elevated NEFAs are associated with inflammation and immune system dysfunction (Bradford et al., 2015; Contreras et al., 2018) and can be a risk factor for ketosis (discussed below).

Plasma glucose concentrations remain stable or increase slightly during the prefresh period, increase dramatically at calving, and then decrease immediately postpartum (Kunz et al., 1985; Vazquez-Anon et al., 1994). The transient increase at calving may result from increased glucagon and glucocorticoid concentrations that promote depletion of hepatic glycogen stores. Although the demand for glucose for lactose synthesis continues after calving, under normal conditions, hepatic glycogen stores begin to replete and are increased by day 14 postpartum (Vazquez-Anon et al., 1994), likely reflecting an increase in gluconeogenic capacity to support lactation. Ketosis can result if this system is dysfunctional (see ketosis section below).

Because of low DMI relative to milk production in the early postpartum period, cows can mobilize substantial amounts of body protein (Bell et al., 2000). Plasma concentrations of 3-methyl histidine (a marker of protein breakdown) can be high the first few weeks of lactation (van der Drift et al., 2012). Estimates of the quantity of body protein mobilized during early lactation vary widely, but most studies show that by about 4 weeks of lactation, body protein mobilization ceases (Komaragiri and Erdman, 1997; Tebbe and Weiss, 2021).

Blood calcium (Ca) decreases the last few days prior to calving due to transfer to colostrum (Goff and Horst, 1997b). Adaptation of the intestine, kidney, and bone to higher demands for Ca takes several days so that blood Ca typically does not return to normal concentrations until several days postpartum. Ca metabolism is discussed in more detail in the hypocalcemia section below.

Ruminal Changes

As a cow transitions from gestation into lactation, the rumen wall undergoes significant changes in size, morphology, and functionality. Because a substantial diet change almost always occurs at the time of calving, diet and physiological changes are confounded. These changes may be caused by metabolic factors, dietary factors, or, more likely, by both. Rumen tissue mass increased about 5 percent between 1 week prepartum and 10 days postpartum (Reynolds et al., 2004) and continued to increase through at least 120 days postcalving (Baldwin et al., 2004; Reynolds et al., 2004). Small intestinal mass followed a similar pattern but reached a peak by 90 days postcalving (Baldwin et al., 2004; Reynolds et al., 2004). Although rumen tissue mass was greater at 10 days postpartum than 7 days prepartum, total rumen volume did not differ (Reynolds et al., 2004). Steele et al. (2015) described several histological differences in rumen epithelial cells collected 3 weeks before compared with 1 week after calving. Postcalving, cells demonstrated accelerated differentiation and desquamation. Morphological differences on the rumen surface are less consistent possibly because of measurement difficulties. In general, rumen papillae size, surface area, and mass increase between late gestation and lactation (Dirksen et al., 1985; Reynolds et al., 2004; Penner et al., 2006; Steele et al., 2015). Expression of genes related to cell growth regulation and differentiation followed patterns that agreed with the histological and morphological changes observed during the transition period (Steele et al., 2015). These changes indicate increasing absorptive capacity within the rumen during the early lactation period.

The immediate postpartum period usually involves a major diet change of reduced concentrations of forage and fiber and increased concentrations of starch and protein, which in addition to increasing DMI will increase concentrations of ruminal volatile FAs and alter their profile. These changes could cause the observed changes in rumen epithelial cells. However, dietary effects on cellular changes occurring at this time are not consistent, and more research is needed. Dirksen et al. (1985) reported increased length and development of ruminal papillae in cows fed a high-concentrate diet postpartum compared with cows fed high-forage diets prepartum; however, diet was completely confounded with all other changes that occur at parturition. Penner et al. (2006) observed that a step-up program in which the amount of concentrate in the diet increased prepartum in a stepwise fashion did not affect papillae growth compared to a high-forage diet prepartum. Reynolds et al. (2004) reported that supplementing 800 g of barley grain to a high-forage diet prepartum increased the number of ruminal papillae per square centimeter of rumen wall but greatly reduced their width and surface area. Andersen et al. (1999) reported no difference in rumen epithelium morphology and development between transition cows fed a high (approximately 85 percent) or low (approximately 45 percent) forage diet. Based on available data, benefits of feeding a diet of moderate starch and fiber to transition ruminal cells and rumen tissue morphology from a high-forage gestation diet to a higher-starch lactation diet are not evident. More research is needed evaluating dietary effects on ruminal morphology and epithelial cell physiology during this critical phase.

Immune Status

During the transition period, cows experience varying degrees of immunosuppression (Goff and Horst, 1997b; Hansen, 2013). Neutrophil and lymphocyte function is depressed (Kehrli et al., 1989a,b; Dosogne et al., 1999; Rinaldi et al., 2008), concentrations of circulating immunoglobulins are reduced at least in part because of transfer to colostrum (Herr et al., 2011), and types and numbers of circulating immune cells are changed (see review by Hansen, 2013). Immunosuppression is one cause for increased prevalence of infectious diseases such as mastitis and metritis that can occur around parturition. Estrogen, progesterone, and cortisol concentrations are high shortly before parturition and can suppress immune function. Nutrient consumption is reduced around parturition, which may also contribute to immunosuppression. Because Ca is integral to immune cell activation (Kimura et al., 2006), hypocalcemia increases the degree of immunosuppression (Martinez et al., 2012). Plasma concentrations of a-tocopherol, retinol, and P-carotene decrease during the periparturient period, and these nutrients can influence immune function (see Chapter 8). Decreases in some of those nutrients can contribute to oxidative stress, which can also contribute to immunosuppression (Sordillo and Aitken, 2009; Sordillo, 2013). Various measures of immune function improved when péripartum cows were fed supplemental rumen-protected methionine (Osorio et al., 2013; Vailati-Riboni et al., 2017). The mode of action is unclear at this time but could involve reduced oxidative stress and inflammation (Han et al., 2018).

Oxidative Stress

Oxidative stress occurs when the relationship between oxidants and antioxidants within a cell or tissue is not appropriate for a specific physiological state. In biological systems, the primary oxidants are reactive oxygen metabolites or species (ROS), but reactive nitrogen species are also important biological oxidants. Free radicals (e.g., superoxide, hydroxyl radical, FA radicals), singlet oxygen, and hydrogen peroxide are major biological ROS. During normal oxidative phosphorylation, a small proportion of the oxygen is not completely reduced to water, resulting in the production of superoxide. Activated phagocytes generate massive quantities of ROS, which are essential for the cells to kill bacteria (Johnston et al., 1975) but contribute to oxidative stress. Cellular antioxidant systems react with ROS and, under normal conditions, maintain proper ROS concentrations. The antioxidant system is composed of enzymes (e.g., superoxide dismutase, catalase, and various glutathione peroxidases), glutathione, a-tocopherol, β-carotene, and ascorbic acid. Superoxide dismutase exists in two forms; the mitochondrial form requires manganese as a cofactor, and the cytosolic form requires copper (Cu) and zinc (Zn). Iron (Fe) is a cofactor for catalase and selenium is a cofactor for glutathione peroxidases. Ascorbic acid is a primary water-soluble antioxidant but can be synthesized by cows and is not considered an essential nutrient for dairy cattle. Adequate dietary supply of antioxidant nutrients can reduce oxidative stress, but in excess, many antioxidant nutrients can act as pro-oxidants and increase oxidative stress (Halliwell, 1996; Lykkesfeldt and Svendsen, 2007).

Oxidative stress can occur under certain disease states (Sordillo and Aitken, 2009; Politis et al., 2012) and is common and may even be normal during the peripartum period (Bernabucci et al., 2005; Castillo et al., 2005; Pedernera et al., 2010; Abueloet al., 2013). Peripartum oxidative stress is higher in cows that are in excess body condition at calving and that undergo greater body condition losses postcalving possibly because of the production of FA radicals from mobilized fat (Bernabucci et al., 2005; Bradford et al., 2015). Oxidative stress can be measured using various biomarkers such as thiobarbituric acid reactive substances (TBARS) and F2-isoprostanes (Celi, 2011; Abuelo et al., 2013) or by measuring concentrations of various antioxidants and ROS (Castillo et al., 2005), but no universally accepted measure or index has been developed. Increased oxidative stress in cattle is associated with mastitis (Politis, 2012; Politis et al., 2012), metritis (Baithalu et al., 2017), edema (Miller et al., 1993), retained placenta (Miller et al., 1993), displaced abomasum (Mudron et al., 1997; Qu et al., 2013), and insulin resistance (Abuelo et al., 2016). Many of these are discussed in more detail in specific sections within this chapter and in Chapters 7 (Minerals) and 8 (Vitamins).

NUTRIENT REQUIREMENTS FOR PREGNANCY AND TRANSITION

Nutrient requirements for pregnancy can be found in the individual nutrient chapters. Colostrum synthesis, although not a gestational requirement, can represent a significant draw on some nutrients during the last several days of gestation. Average first-milking colostrum yields are about 4 kg (first lactation) to 8 kg (older cows), and first-day colostrum yields range from 8 to 14 kg for Holsteins (Bobe et al., 2009; Kessler et al., 2014). Average first-milking colostrum yield by Jerseys was about 4 kg, but season (approximately 2.5 kg in winter and 6.6 kg in summer) had a large effect (Gavin et al., 2018). Average concentrations of some important nutrients in colostrum are in Table 12-1 (Cu, manganese, and Fe are in trace concentrations and not shown).

Assuming a yield of 10 kg of colostrum and average concentrations of nutrients (see Table 12-1), the average Holstein cow would secrete 14 Meal of energy, 1.35 kg of protein, 21 g of Ca, 30 mg of retinol, and 60 mg of tocopherol in her

TABLE 12-1 Average Composition of Colostrum[a]

Nutrient	Concentration	Reference
Crude protein, g/kg	145 (193)[b]	Nocek et al., 1984; Quigley et al., 1994; Hammon et al., 2000; Bobe et al., 2008
Fat, g/kg	65 (50)	Hammon et al., 2000; Bobe et al., 2008
Lactose, g/kg	25 (25)	Bobe et al., 2008
Gross energy, Mcal/kg	1.4 (1.5)	Hammon et al., 2000
Retinol, mg/kg	3.0	Johnston and Chew, 1984; Puvogel et al,, 2008
a-Tocopherol, mg/kg	6.0	Weiss et al., 1990b, 1997; Rajaraman et al., 1997; Kumagal and Chaipan, 2004
Vitamin B_{12}, gg/kg	19	Stemme et al,, 2006
Calcium, mg/kg	2.1 (2.4)	Foley and Otterby, 1978; Roux et al., 1979; Salih et al., 1987; Shappell et al,, 1987; Kume and Tanabe, 1993; Tsioulpas et al., 2007
Phosphorus, mg/kg	1.8	Salih et al., 1987; Shappell et al., 1987; Kume and Tanabe, 1993; Tsioulpas et al., 2007
Magnesium, mg/kg	0.3	Foley and Otterby, 1978; Salih et al,, 1987; Shappell et al., 1987; Kume and Tanabe, 1993; Tsioulpas et al., 2007
Potassium, mg/kg	1.3	Salih et al,, 1987; Kume and Tanabe, 1993; Tsioulpas et al., 2007
Sodium, mg/kg	0.9	Foley and Otterby, 1978; Shappell et al., 1987; Kume and Tanabe, 1993; Tsioulpas et al., 2007
Copper, mg/kg	0.5	Moeini et al,, 2011
Selenium, mg/kg	0.04 to 0.17[c]	Weiss and Hogan, 2005; Salman et al., 2013
Zinc, mg/kg	15	Foley and Otterby, 1978; Vaillancourt and Allen, 1991; Moeini et al., 2011

[a] Data are from Holstein cows unless otherwise noted.
[b] Values in parentheses are for Jersey cows.
[c] Low value represents colostrum from cows fed inorganic Se and high value is from cows fed selenium-yeast.

colostrum the first day postpartum. The impact of Ca secretion is discussed in the hypocalcemia section. The secretion of protein into colostrum is not included in the requirements for transition cows because it is short term, and the preponderance of data does not show any beneficial effects of increased protein supply during the prepartum period (discussed below). However, data on the effects of prepartum protein supply on colostrum yield, nutrient composition, and overall quality are essentially nonexistent. The secretion of retinol and tocopherol into colostrum clearly affects vitamin A and vitamin E status of the cow. Plasma concentrations of retinol and tocopherol drop abruptly about 1 week before parturition (Goff and Stabel, 1990; Weiss et al., 1990a), and a large portion of that decrease is because of transfer to colostrum (Goff et al., 2002). Increasing the intake of vitamin E from 1,000 IU/d (approximate requirement) to 4,000 IU/d during the last 2 weeks of gestation prevented the decrease in plasma concentrations of tocopherol and reduced clinical mastitis (Weiss et al., 1997). Feeding 2,000 IU/d of supplemental vitamin E, but not 1,000 IU/d during the last 2 weeks of gestation, attenuated the decrease in plasma tocopherol concentrations around parturition and reduced somatic cell count (Baldi et al., 2000). Conversely, the decrease in plasma retinol occurs even when cows are fed very high concentrations of vitamin A. Cows fed 550,000 IU/d of supplemental vitamin A (approximately seven times the 2001 NRC requirement) during the dry period still exhibited a decrease in plasma retinol around calving. Assuming a first-day colostrum yield of 10 kg, a cow will lose about 100,000 IU of vitamin A (30 mg of retinol) and about 135 IU of vitamin E (50 mg of /?////-tocopherol) via colostrum the first day after parturition. Colostrum synthesis occurs during the last few days of gestation; therefore, during most of the dry or prefresh period, no increased demand for colostral vitamins A and E exists. Hence, the loss of nutrients via colostrum is not included in requirement calculations, but users should recognize the potential impact of colostrum synthesis on overall nutrient requirements of the very late-gestating cow or heifer.

Dry Matter Intake During the Dry Period

Factors Affecting Dry Matter Intake

Feed intake is relatively constant during the initial phase of the dry period (days 60 to 21 prepartum) but can decline quite dramatically thereafter, especially during the 7 to 10 days prior to calving (Hayirli et al., 2003). The major animal factors that influence DMI during this time are body weight (BW), day of gestation, parity, body condition, and health (Grummer et al., 2004; Hayirli and Grummer, 2004). Prefresh cows with excess body condition (>4 on a 5-point scale) consumed about 8 percent less dry matter (DM) than cows at similar BW but with lower body condition scores (Hayirli et al., 2002). Cows with developing metabolic or health problems, including hypocalcemia, have lower DMI prepartum than healthy cows (Goff and Horst, 1997b: Huzzey et al., 2007).

Increasing dietary energy (Coppock et al., 1972; Hernandez-Urdaneta et al., 1976; Minor et al., 1998) or dietary energy and protein (VandeHaar et al., 1999) concentrations during the prefresh period resulted in higher DM and energy intake. However, concentration of dietary crude protein (CP) within the range of about 10 to 16 percent generally does not affect intake in dry and prefresh cows (Hayirli et al., 2002). Concentrations of rumen-undegradable protein (RUP) were negatively associated with prepartum intake, but that may be more related to source of protein (i.e., most of the studies with high RUP fed animal-based proteins) rather than RUP per se. Increasing dietary fat concentration may reduce DMI, but that effect was only observed in prepartum heifers (Hayirli et al., 2002).

Similar to lactating cows (see Chapter 2), diet digestibility, as reflected by source and concentration of neutral detergent fiber (NDF) or estimated net energy for lactation (NEL), is probably the most important dietary factor related to DMI by dry cows. Increasing the NDF concentration of dry cow or prepartum heifer diets by the addition of lower-quality grass (Holcomb et al., 2001), cottonseed hulls (VandeHaar et al.. 1999), or straw (Dann et al., 2006; Janovick and Drackley. 2010; Mann et al., 2015) usually reduces DMI. Hayirli et al. (2002) classified diets from multiple studies into three groups: low NDF (28 to 32 percent), medium NDF (35 to 50 percent), and high NDF (50 to 62 percent). Average (across parities) daily DMI for those three classes of diets by animals during the last 3 weeks of gestation was approximately 2.0, 1.7. and 1.6 kg per 100 kg of BW (Hayirli et al., 2002). If the NDF is more digestible (e.g., from soyhulls), DMI can be much higher than those values (Holcomb et al., 2001). An interaction between day of gestation and NDF on DMI may also occur (Janovick et al., 2011). In most studies cited above, as cows approached parturition, the decrease in DMI was less when cows were fed diets that included lower-quality feeds such as straw or mature grass than the control diets. The effect of parity (independent of BW) on DMI during the last 1 to 2 months of gestation is not clear. Using data from multiple studies, on a BW basis, nulliparious animals consumed about 12 percent less DM than multiparious cows during that period (Hayirli et al., 2003). However, Janovick et al. (2011) reported that from about 35 to 21 days prepartum, daily DMI of heifers and cows when fed a higher digestible diet (39 percent NDF) was similar (approximately 2 kg of DMI/100 kg of BW). When animals were fed a straw-based diet (51 percent total NDF), DMI by heifers was about 25 percent less on a BW' basis than intake by cows (Janovick et al., 2011).

Predicting Dry Matter Intake in Transition Cows and Dry Cows

In the previous edition, daily DMI (as kg/100 kg of BW) during the last 3 weeks of gestation was estimated using loga-

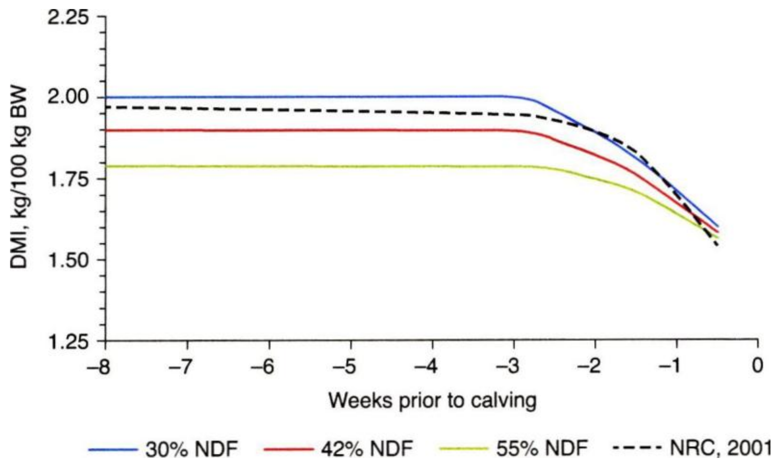

FIGURE 12-1 Estimated (Equation 12-1 and NRC, 2001) daily DMI by cows (>1 parturition) fed diets with 30 to 55 percent NDF (mostly from forages) during the dry period.

rithmic decay functions (one for heifers and one for cows) based solely on day relative to calving (Hayirli et al., 2003). The estimated daily DMIs between 60 and 22 days prepartum using those equations were 1.7 and 2.0 kg DM/100 kg BW for heifers and cows, respectively. Based on those equations, DMI started to decrease sooner (relative to calving) for cows than for heifers so that by calving, DMI per unit of BW was similar between parities. Based on limited data, the model was accurate for heifers but was inaccurate (mean square prediction error was 77 percent of mean intake) and biased (mean overestimation of 1.1 kg) for cows. The low accuracy is at least partially attributed to the lack of any dietary factors in the equation (Hayirli et al., 2002).

Inadequate data were available to conduct a true meta-analysis. Data used by Hayirli et al. (2003) plus recently published data (Holcomb et al., 2001; Dann et al., 2006; Janovick and Drackley, 2010; Mann et al., 2015) were combined to estimate DMI based on diet NDF and stage of gestation using multiple regression. The concentration of forage or roughage NDF was not reported in most studies, and only studies in which forage was the predominant source of NDF were used. Predicted intake is likely inaccurate when cows are fed diets that contain substantial amounts of by-products with more digestible NDF (e.g., soyhulls or distillers grains). The following equation is valid only for diets between 30 and 55 percent NDF. The equation to estimate DMI during the last 3 weeks of gestation for cows is as follows:

$$\text{Daily DMI, kg/100 kg BW} = 1.47 - [(0.365 - 0.0028 \times \text{NDF}) \times \text{Week}] - 0.035 \times \text{Week}^2 \quad \text{(Equation 12-1)}$$

w here NDF is percentage of diet DM (assumed to be mostly from forage). If NDF <30 percent, then NDF = 30, and if NDF >55 percent, then NDF = 55. Week is week prior to calving entered as a negative number. Limited data suggest that for cows with body condition score (BCS) >4, estimated DMI should be reduced by 8 percent (Hayirli et al., 2003).

DMI during the early phase of the dry period was set at the same value as for week 3 (e.g., 2.0, 1.9, and 1.8 kg/100 kg of BW for cows fed diets with 30, 40, or 50 percent NDF, respectively). Based on Equation 12-1, during the early dry period, reducing dietary NDF increases DMI, but as cows approach parturition, the effect of NDF becomes less (see Figure 12-1).

Because of inadequate data, no interactions with parity, NDF, and stage of gestation could be modeled. Therefore, the same DMI equation for cows (see Equation 12-1) is used for heifers, except DMI is reduced by 12 percent, which is the difference in intake between heifers and cows in the early phase of the dry period (60 to 22 days prepartum) observed by Hayirli et al. (2003).

$$\text{Heifer DMI, kg/100 kg BW} = (1.47 - [(0.365 - 0.0028 \times \text{NDF}) \times \text{Week}] - 0.035 \times \text{Week}^2) \times 0.88 \quad \text{(Equation 12-2)}$$

Rather than estimating DMI based on day before calving, which will not be known until after the animal has calved, time is expressed in weeks to introduce a degree of uncertainty (3 weeks before calving encompasses the time between 21 and 15 days before calving, 2 weeks is 14 to 8 days prepartum, and 1 week is between 7 days and 1 day prepartum). These equations and values were derived using data from Holstein cattle and are assumed to work for other breeds.

Effect of Prepartum Diet on Postpartum Production

Protein

Based on estimated protein use (i.e., maintenance, fetal growth, mammary gland development), cows and heifers may have to mobilize body tissue during the immediate pre- and postpartum period to meet those needs. However, direct measurements of changes in maternal protein reserves during late gestation are lacking. Putnam and Varga (1998) measured apparent nitrogen balance in multiparous cows at approximately 10 days prepartum. Cows were fed diets containing 10.6, 12.7, and 14.5 percent CP; DMI was not affected (11.3 kg/d), but as dietary CP concentration increased, apparent nitrogen retention increased from 36 to 49 g/d. Those values include nitrogen retained in the fetus. After correcting for estimated fetal protein deposition, maternal nitrogen balance was still positive for all treatments, suggesting that even a diet containing 10.6 percent CP was adequate to maintain maternal protein stores. Comparable data for heifers are lacking. Because heifers consume less feed and have greater demands for mammary gland development and growth than cows, changes in maternal protein reserves may differ between prepartum cows and heifers.

Because of the potential for losses of maternal protein stores in late gestation, several studies (see Lean et al., 2013) have evaluated the effects of increasing the concentration of CP in prepartum diets on postpartum production. Across those studies, CP concentration of the low-protein diets averaged approximately 12 percent (range from 10 to 13), and the average concentration of the high-protein diets averaged approximately 16 percent (range from 12 to 23 percent). Some individual studies reported increased yields of milk or milk protein when higher CP diets were fed prepartum (Santos et al., 2001), but others reported negative effects of increased prepartum CP (Greenfield et al., 2000). A meta-analysis determined that on average, dietary CP concentration prepartum had no effect on milk yield (Lean et al., 2013). Likewise, providing additional RUP prepartum has not consistently affected milk production postpartum. Studies to determine effects of supplemental amino acids during the prepartum period independent of postpartum supplementation on postpartum production are lacking. Feeding diets with more than about 12 percent CP to cows during the immediate prepartum period will likely not increase milk production in the subsequent lactation. Because of lower DMI and potentially greater requirements, heifers may benefit from higher dietary concentrations of CP prepartum, but data are lacking. A meta-analysis found that postpartum production by cows was not affected by prepartum metabolizable protein (MP) supply, but increasing prepartum MP supply to heifers was beneficial (Husnain and Santos, 2019). The effect of prepartum protein on colostrum quality and yield has not been evaluated. In addition, data are lacking on the effect of varying dietary CP or MP concentration when prepartum cows are fed low-energy (e.g., high-straw) diets.

Protein requirements for mammary growth were not included in the computer model mainly because insufficient data for mammary parenchymal growth rates and the composition of that growth are available. VandeHaar and Donkin (1999) estimated that the additional CP for mammary growth during the last few weeks of gestation would be approximately 120 to 130 g/d, which is equivalent to about 1 percentage unit of dietary CP, assuming mammary parenchymal mass increased by 430 to 460 g/d during the transition period (NRC, 2001). The meta-analysis by Husnain and Santos (2019) supports feeding higher dietary protein prepartum to heifers.

Energy and Carbohydrates

Energy density and concentrations of NDF and starch in prepartum diets are highly correlated (e.g., increasing NDF concentration usually reduces starch and the NEL concentrations). Therefore, effects of changing one of those components cannot be isolated, and all three will be discussed concurrently.

Several studies have examined how energy intake during the prepartum period affects postpartum performance and health (Kunz et al., 1985; Holter et al., 1990; Olsson et al.. 1998; Mashek and Beede, 2001; Agenas et al., 2003; McNamara et al., 2003; Dann et al., 2006; Silva-del-RIo et al.. 2010; Mann et al., 2015). Generally, one treatment consisted of prepartum energy intake at approximately the cow’s requirement, and the other treatment or treatments consisted of substantially greater energy intake (e.g., 25 to 50 percent excess energy intake). Energy and DMI was usually increased by replacing lower-quality forage NDF with starch or higher-quality forages. Postpartum intakes usually were not affected by prepartum treatment (cows fed a common diet after calving). In some studies (Janovick and Drackley. 2010; Silva-del-RIo et al., 2010; Richards, 2011), milk production was lowered by reducing energy intake prepartum. but in most of the studies, milk production was not affected by prepartum energy intake.

A common practice is to formulate higher-energy diets for closeup or prefresh cows (i.e., cows during the last 2 to 3 weeks prepartum) by increasing the concentration of starch and reducing the amount of forage and fiber in the diet. Potential benefits are to aid in the adaptation of rumen microbes to the higher-starch diet that will be fed after calving and to maintain DM and energy intake during the immediate prepartum period. The rumen microbial population changes as a cow transitions from prepartum to postpartum (Minuti et al., 2015) likely because of dietary changes. Transitioning growing beef cattle from a forage-based diet (ca. 50 percent forage) to a high-concentrate diet (ca. 90 percent) over a period of 2 weeks compared with an abrupt change resulted in greater average daily gains or better feed efficiency, and the ruminal microbial populations required days to weeks to stabilize during the diet change (Brown et al., 2006). The dietary changes a typical dairy cow undergoes during the transition period are substantially less dramatic, but data are lacking on how transition diets may affect ruminal microbial populations.

Increasing starch and reducing fiber concentrations in prefresh diets usually increases DM and energy intakes during the prefresh period, but this usually does not translate into greater postpartum intakes or milk yields (Minor et al., 1998:

Holcomb et al., 2001; Keady et al., 2001; Doepel et al., 2002; Rabelo et al., 2003, 2005; Smith et al., 2005; Vickers et al., 2013) . Even when prefresh feed intake was restricted by almost 50 percent, postpartum intake and milk production were not affected (Dann et al., 2006). Evidence suggests that prefresh diets with lower-energy concentrations reduce the risk of subclinical and clinical ketosis during the subsequent lactation (Doepel et al., 2004; Smith et al., 2008; Vickers et al., 2013).

ETIOLOGY AND NUTRITIONAL PREVENTION OF METABOLIC DISORDERS

Fatty Liver

Fat accumulation in the liver occurs when uptake of FAs exceeds the capacity of the liver to oxidize or secrete the FAs, which happens when blood concentrations of NEFAs are elevated (Bobe et al., 2004; Grummer, 2008). Ketosis almost always occurs when cows have moderate (5 to 10 percent of liver wet weight as triacylglyceride) to severe (greater than 10 percent fat) fatty liver. In healthy cows, plasma NEFA concentrations are low (less than about 0.2 mEq/F) until a few days before parturition and then reach concentrations as high as 0.8 mEq/L at calving, remain high for a few days, and then start slowly decreasing (Bertics et al., 1992; Grum et al., 1996; LeBlanc et ah, 2005). Cows that are at greater risk to develop metabolic problems postpartum often have plasma NEFA concentrations greater than 0.5 mEq/F prepartum and can be much greater than 1.0 mEq/L postpartum (LeBlanc et ah, 2005; Roberts et ah, 2012). Plasma NEFAs are elevated when cows are in negative energy balance, which can occur when DMI drops during the immediate prepartum period and almost always occurs during the first few weeks of lactation because intake is low and energy needs are high, causing mobilization of body reserves. Because most cows have elevated NEFAs during the peripartum period, most early lactation cows have some degree of hepatic fat accumulation (Jorritsma et ah, 2001; Bobe et ah, 2004).

Extensive reviews of the regulation of hepatic lipid metabolism and its relation to fatty liver and ketosis have been published (Emery et ah, 1992; Grummer, 1993; Drackley, 1999; Hocquette and Bauchart, 1999; White, 2015). Uptake of NEFAs by the liver is proportional to NEFA concentrations in blood (Emery et ah, 1992), and NEFAs taken up by the liver can be esterified or oxidized (Drackley, 1999). The primary esterification product is triglyceride (TG), which can be exported as part of a very low-density lipoprotein or be stored. In ruminants, export of TG occurs at a very slow rate relative to other species (Kleppe et ah, 1988; Pullen et ah, 1990). Therefore, under conditions of elevated hepatic NEFA uptake, FA esterification and TG accumulation occur.

Fatty liver is a major risk factor for displaced abomasum, ketosis, and immune dysfunction (Bobe et ah, 2004). Conversely, these disorders may be a risk factor for fatty liver if they reduce DMI, causing a more severe negative energy balance. In addition to clinical abnormalities, fatty liver is associated with reduced function by hepatocytes (e.g., reduced gluconeogenesis and reduced ureagenesis), increased measures of oxidative stress (e.g., lower concentrations of plasma a-tocopherol and higher concentrations of malondialdehyde), and increased inflammation (Bobe et ah, 2004; Bradford et ah, 2015).

A major risk factor for development of fatty liver is obesity (Bobe et ah, 2004; Roche et ah, 2013). During the early postpartum period, fat cows usually have lower DMI than cows in proper body condition, resulting in greater mobilization of body fat (Stockdale, 2001). More severe negative energy balances in early lactation were associated with greater concentrations of hepatic lipid and plasma NEFA (Weber et ah, 2013) concentrations of blood NEFAs. Prepartum intake per se does not appear to be related to fatty liver; however, Grummer (2008) hypothesized that prepartum change in intake or, more specifically, change in energy balance may be a cause of hepatic fat accumulation. Increasing dietary starch concentration during the prefresh period usually does not reduce liver fat accumulation postpartum, even though it often increases DMI in the prepartum period and may reduce NEFA concentrations around calving (Overton and Waldron, 2004; Grummer, 2008). Increased dietary starch concentrations in the immediate postpartum period have reduced hepatic fat accumulation (Rabelo et ah, 2005). Increasing the concentration of dietary fat during the peripartum period has not consistently affected liver fat accumulation (Skaar et ah, 1989; Bertics and Grummer, 1999; Andersen et ah, 2008). Responses may be affected by concentration of added fat and type of FAs (e.g., saturated FAs may decrease hepatic lipid concentrations [Andersen et ah, 2008]). Monensin often reduces blood ketones, but it has not been shown to reduce hepatic fat accumulation (Zahra et ah, 2006; Duffield et ah, 2008). Supplementing rumen-protected choline during the peripartum period can reduce liver fat concentrations (Cooke et ah, 2007; Zom et ah, 2011; Lima et ah, 2012; Elek et ah, 2013; Zenobi et ah, 2018). Niacin has antilipolytic properties, but unless supplemented at very high rates, it usually does not affect plasma NEFA concentrations (reviewed by Grummer, 2008). Supplementing peripartum cows with rumen-protected niacin has reduced plasma NEFAs but has not markedly affected liver lipid concentrations (Yuan et ah, 2012; Morey et ah, 2011).

Ketosis

Ketosis or hyperketonemia occurs when excessive amounts of long-chain FAs are oxidized via (3-oxidation. Excessive amounts of long-chain FAs are released in cows undergoing severe negative energy balance after parturition. Ketone bodies (β-hydroxybutyric acid and acetoacetate) are end products of (3-oxidation, and when these accumulate in the blood, clinical signs can be observed. Release of FAs from adipose tissue followed by (3-oxidation is stimulated when plasma insulin is low and glucagon is high (Holtenius and Holtenius, 1996). Many of the clinical signs of ketosis such as reduced DMI and milk production and lethargy are nonspecific. Some cows will show abnormal behavior such as aggression, incoordination, and chewing on nonfood objects. A definitive diagnosis requires some measure of ketones in blood, urine, or milk. The accuracy and value of various tests have been reviewed (Tatone et al., 2016) and will not be discussed. For this discussion, (i-hydroxybutyric acid (BHBA) in blood will be used as the standard diagnostic for ketosis. Elevated blood BHBA has been associated with increased risk of numerous health problems, reduced milk yield, and reduced reproductive efficiency (Walsh et al., 2007; Ospina et al., 2010; Chapinal et al., 2012; Suthar et al., 2013; Raboisson et al., 2014). The cutoff for separating healthy lactating cows from cows with subclinical ketosis has varied between approximately 1.0 and 1.4 mmol/L (Duffield, 2000; Raboisson et al., 2014), but a value of >1.2 mmol/L of BHBA is commonly used to define subclinical ketosis. Incidence rates for subclinical ketosis will depend on which cutoff value is used, timing of blood sampling, and so on, but herd-level rates of 20 to 40 percent have been reported (Duffield et al., 2009; McArt et al., 2012; Suthar et al., 2013).

Holtenius and Holtenius (1996) classified ketosis as either type 1 or type 2. Type 1 ketosis generally occurs a few weeks after parturition when milk production and glucose demand by the mammary gland are high and is usually not associated with excessive hepatic fat concentrations. Type 2 occurs at or very near parturition and is usually associated with fatty liver. Type 2 ketosis is often more refractory to treatment than type 1 (Herdt, 2000). This classification scheme illustrates the two major causes of ketosis. Risk factors and causes of type 2 ketosis are largely the same as those for fatty liver (discussed above). With type 1, blood glucose and insulin concentrations are lower and ketone concentrations are higher compared to healthy cows. Low insulin probably enhances FA oxidation by decreasing hepatocyte malonyl-CoA concentrations and sensitivity of carnitine palmitoyltransferase 1 to malonyl-CoA concentrations (Emery et al., 1992). Carnitine palmitoyltransferase 1 is responsible for translocating FAs from the cytosol to the mitochondria for oxidation, and its activity is high with type 1 ketosis. This suggests that for type 1, the supply of precursors for gluconeogenesis is not adequate. Limited availability of substrate could be caused by low DMI. Increasing dietary starch postpartum reduces blood BHBA and increases glucose (Rabelo et al., 2005; McCarthy et al., 2015). Supplementing monensin postpartum reduces blood BHBA and increases glucose (Sauer et al., 1989; McCarthy et al., 2015). Administration of propylene glycol as either a drench or a bolus consumption can reduce blood BHBA (Nielsen and Ingvartsen, 2004).

Udder Edema

Udder edema is characterized by excessive accumulation of fluids in the intercellular tissue spaces of the mammary gland usually during the péripartum period. Edema and congestion occur in the udder and umbilical area and may be prominent in the vulva and brisket. Incidence rate in periparturient Holstein heifers was about 12 percent in one herd in Florida (Melendez et al., 2006); however, Morrison et al. (2018) reported a 66 percent incidence rate in three herds in Ontario. Case descriptions were not the same for both studies. Typically, the incidence and severity of udder edema are greater in pregnant heifers than in cows (Zamet et al., 1979; Erb and Grohn, 1988) and tend to be more severe in older than in younger heifers and in heifers with male calves rather than female calves (Melendez et al., 2006). Obese cows (Vigue, 1963) and cows that had udder edema previously (Melendez et al., 2006) are at increased risk for udder edema. Udder edema is moderately heritable (Dentine and McDaniel, 1983). Edema can be a major discomfort to the animal and causes difficulty with milking machine attachment, increased risk of teat and udder injury, and mastitis. Severe udder edema may reduce milk production and cause a pendulous udder (Dentine and McDaniel, 1983). The exact cause(s) of udder edema is unknown; more likely, it is a multifactorial condition. Restriction or stasis of venous and lymph flow from the udder in late pregnancy due to fetal pressure in the pelvic cavity causing increased venous pressure may be a contributing factor (Vestweber and Al-Ani, 1983, 1984; Al-Ani and Vestweber, 1986). Changes in amounts and relative proportions of steroid hormones during late pregnancy may also be involved. Reduced concentrations of proteins, especially globulins, in blood suggest an increase in vascular permeability as animals approach calving and have been associated with greater incidences of udder edema (Vestweber and Al-Ani, 1984).

Emery et al. (1969) reported increased udder edema in heifers fed high-concentrate diets, but that may have been caused by the approximate 75-g/d increase in sodium chloride (NaCl) intake rather than the concentrate per se. Excessive intakes of sodium (Na) and potassium (K) have been implicated as causative agents in udder edema (Randall et al., 1974; Sanders and Sanders, 1981; Vestweber and Al-Ani, 1983; Al-Ani and Vestweber, 1986). Restriction of NaCl and water intakes reduced the severity and incidence of udder edema in pregnant heifers (Hemken et al., 1969). Lower incidence and severity of udder edema were found when diets contained no supplemental salts of Na or K (Randall et al., 1974). In a field study with two commercial dairy herds, K fertilization of alfalfa was implicated as the cause of increased udder edema (Sanders and Sanders, 1981). Cows consumed about 450 g of K/head per day. In an earlier controlled study, consumption of 454 g of a combination of NaCl and potassium chloride (KC1) increased the incidence and severity of udder edema (Randall et al., 1974). In a second study, the incidence and severity of udder edema in pregnant heifers fed a grain mix containing 1 percent NaCl or a grain mix with 4 percent supplemental KC1 plus 1 percent NaCl for 20 days did not differ (Randall et al., 1974). Nestor et al. (1988) reported that the severity of udder edema was greater when pregnant heifers were fed sodium bicarbonate (0 versus 272 g/head per day) or NaCl (23 versus 136 g/head per day) separately but not when both salts were fed together. Excessive intake of the chloride salts of Na or K probably increases the severity of udder edema, especially in late-pregnant heifers. Using forages with low concentrations of K and limiting supplemental Na would be prudent if udder edema is prevalent.

Lema et al. (1992) and Tucker et al. (1992) studied the effects of prepartum calcium chloride ($CaCl_2$), an anionic salt, on the incidence and severity of udder edema. Results were mixed, but supplementation of $CaCl_2$ at approximately 1.5 percent of diet DM often reduced the prevalence and severity of edema in the peripartum period. Oxidative stress may play a role in udder edema (Miller et al., 1993; Mueller et al., 1998). Mueller et al. (1998) reviewed two studies on the effects of antioxidant and pro-oxidants on udder edema. In one study, udder edema during the first week after calving was less in heifers supplemented for 6 weeks before calving with 1,000 IU vitamin E/head per day versus none. In the other study, late-pregnant heifers were fed factorial combinations of vitamin E (0 or 1,000 IU/d), Zn (0 or 800 mg/d), and Fe from iron sulfate (0 or 12 g/d, which is equal to about 1,300 mg/kg of diet DM). Fe can be a pro-oxidant and increases the formation of ROS. Without supplemental Fe, vitamin E reduced severity of udder edema, but Zn did not. When Fe was excessive, vitamin E was ineffective in reducing the severity of udder edema, but Zn was somewhat effective, perhaps by reducing absorption of Fe. This suggests that when ROS are extremely high (e.g., high concentrations of reduced Fe in the diet), antioxidants may not be able to overcome their effects.

Retained Placenta and Metritis

Retained placenta (retained fetal membranes) is defined as failure of the fetal membranes to be expelled within 24 hours after parturition (Kelton et al., 1998). Metritis is defined as postpartum cows with abnormally enlarged uterus with fetid red-brown watery or purulent vaginal discharge within the first 21 days after calving with or without systemic signs of illness (e.g., fever). Most cows (60 to 80 percent) with retained placenta will have metritis, but the incidence of metritis is usually much greater than the incidence of retained placenta (Gilbert et al., 2005; Han and Kim, 2005). Retained placenta and metritis impair various measures of reproductive efficiency (Erb et al., 1985; Opsomer et al., 2000; Giuliodori et al., 2013). Cows that had a retained placenta usually have reduced milk yields (Joosten et al., 1988; Rajala and Grohn, 1998; Gilbert et al., 2005), but effects of metritis on subsequent milk production are less clear (Giuliodori et al., 2013) possibly because of the varied definitions and severity of metritis.

Multiple physiologic and nutritional factors have been implicated as causes of retained placenta and metritis. Dystocia, twinning, stillbirth, and caesarean section increase the risk for retained placenta (Erb et al., 1985; Han and Kim, 2005). However, cows that had a retained placenta had elevated serum concentrations of several proinflammatory cytokines and lactate as early as 8 weeks prepartum, indicating calving events are not the only cause of retained placenta (Dervishi et al., 2016). Older cows generally are at greater risk than first-parity cows, and a short gestation period increases risk of retained placenta (Bendixen et al., 1987; Grohn et al., 1990; Han and Kim, 2005).

Nutritional Factors

Most studies evaluating nutritional influences on retained placenta evaluate supplementation during the entire dry period or during the last 2 or 3 weeks of gestation. Inadequate supply of selenium (Se), vitamin E, vitamin A, and |3-carotene is related to increased prevalence of retained placenta. Lower concentrations of serum Zn are associated with increased retained placenta (Sheetal et al., 2014), but data showing that supplementation of Zn reduces retained placenta are lacking. Many or perhaps all of these effects could be mediated via improved immune function. Immune system dysfunction, specifically reduced neutrophil function, prior to parturition was associated with increased prevalence of retained placenta (Kimura et al., 2002). Supplementing Cu to Cu-deficient cows improved the function of neutrophils (Torre et al., 1996); however, clinical data on effects of Cu supplementation on retained placenta are lacking. These nutrients are also involved in cellular antioxidant systems, and maintaining proper concentrations of ROS within cells and tissue can affect production of various prostaglandins and eicosanoids, which can affect placental retention. Oxidative stress increases around parturition, and cows that demonstrate increases in oxidative stress earlier in the prepartum period, to a greater extent, have more severe metritis postpartum (Baithalu et al., 2017) and are more likely to have a retained placenta (Miller et al., 1993).

An excess of ROS can cause peroxidative damage of cell membranes and interfere with normal metabolic function, including normal steroidogenesis (Miller et al., 1993) and arachidonic acid metabolism (Sordillo, 2013). Supplementing diets with antioxidants to meet requirements is crucial during the periparturient period (Weiss et al., 1990a) when blood a-tocopherol concentrations are the lowest of the entire lactation cycle (Goff and Stabel, 1990; Weiss et al., 1990a), and expression of several antioxidant enzymes (Aitken et al., 2009) and total antioxidant capacity (Castillo et al., 2005; Baithalu et al., 2017) is low. Based on the preponderance of data, when basal diets with <0.1 mg of Se/kg of diet DM are supplemented with an additional 0.1 to 0.3 mg/kg Se or cows are injected with Se approximately 2 to 3 weeks prepartum, prevalence of retained placenta is decreased (Trinderet al., 1969; Julien et al., 1976a,b; Segerson et al., 1981; Harrison etal., 1984; Jovanovic et al., 2013). Se supplementation did not always reduce retained placenta, but in most of those instances, basal diets contained more than 0.1 mg Se/kg of DM (Gwazdauskas et al., 1979;

Schingoethe et al., 1982; Hidiroglou et al., 1987; Stowe et al., 1988). Vitamin E supplementation also significantly reduces the risk of retained placenta (Bourne et al., 2007), and the concentration of serum α-tocopherol prepartum was lower in cows that went on to have a retained placenta compared with healthy cows (Qu et al., 2014). The recommended intakes for Se (see Chapter 7) and vitamin E (see Chapter 8) were derived in part from experiments evaluating their effects on retained placenta, and increasing intakes above recommended values will likely not affect prevalence of retained placenta.

Cows showing clinical signs of vitamin A deficiency had increased incidence of retained placenta (Nicholson and Cunningham, 1965). However, in cows with better vitamin A status, no relationship was observed between serum retinol concentrations and retained placenta (LeBlanc et al., 2004). Supplemental (3-carotene at 600 or 1,200 mg/d has reduced incidence of retained placenta (Michal et al., 1994; Oliveira et al., 2015). However, in the Oliveira et al. (2015) study, supplementation only reduced prevalence in multiparous cows. LeBlanc et al. (2004) found no difference in serum (3-carotene concentrations between late-gestation cows that eventually developed retained placenta and those that did not. Based on available data, after meeting vitamin A recommendations (see Chapter 8), additional supplementation is not expected to affect prevalence of retained placenta. Although (3-carotene supplementation can reduce prevalence of retained placenta, inadequate data are currently available to derive an Adequate Intake value.

Cows with hypocalcemia have a higher risk for retained placenta than cows with normal blood Ca (Curtis et al., 1985; Rodriguez et al., 2017), and factors related to hypocalcemia are discussed below. Supplementing late-gestation cows with calcidiol (25-OH vitamin D_3) rather than cholecalciferol (vitamin D_3) greatly reduced the incidence of retained placenta and metritis (Martinez et al., 2018). Vitamin D is related to immune function; however, in that experiment, calcidiol supplementation did not affect neutrophil function prepartum.

Extreme deficiencies of energy, protein, or both can result in retained placenta because cows are weak and, coupled with the stress of parturition, lack strength to expel the placenta (Maas, 1982). Cows fed diets low in CP (8 percent) for the entire dry period had a higher incidence of retained placenta compared with cows fed 15 percent CP (50 versus 20 percent incidence) (Julien et al., 1976a). Fat cows (Morrow, 1976) and cows with elevated plasma NEFAs or ketones prepartum have a higher risk of retained placenta and metritis (Qu et al., 2014; Raboisson et al., 2014).

Milk Fever (Hypocalcemia)

Plasma concentrations of Ca should be 9 to 10 mg/dL (2.25 to 2.5 mM). However, an acute and severe form of hypocalcemia known as milk fever occurs in nearly 5 percent of multiparous dairy cows as a result of the large and sudden secretion of Ca in milk that occurs at the onset of lactation (NAHMS-USDA, 2018). Blood Ca concentrations are often below 4.5 mg/dL (1.12 mM) in these recumbent cows exhibiting muscle paresis. About 50 percent of multiparous dairy cows and 25 percent of heifers experience a subclinical hypocalcemia around the time of calving (Reinhardt et al., 2011). Cows with plasma concentrations less than 8.0 (Reinhardt et al., 2011) to 8.6 mg/dL (Martinez et al., 2012) are considered subclinically hypocalcémie. Cows with subclinical hypocalcemia are at increased risk for immune dysfunction, metritis, displacement of the abomasum, retained placenta, mastitis, and ketosis (Daniel, 1983; Massey et al., 1993; Kimura et al., 2006; Martinez et al., 2012; Chamberlin et al., 2013; Neves et al., 2018; McArt and Neves, 2020). However, time of sampling relative to calving has a marked effect on plasma concentrations of Ca, and the health risks associated with low plasma Ca depend on when (relative to calving) the sample was taken (Neves et al., 2018; McArt et al., 2020). For example, low plasma Ca 1 day postpartum was not related to increased risk of cows developing health problems, but cows with low plasma Ca on day 2,3, or 4 postpartum had increased risk for metritis and displaced abomasum (Neves et al., 2018). Furthermore, the concentration of plasma Ca at which a statistically increased risk of other health disorders occurs differs between primiparous and multiparous cows (Neves et al., 2018).

Hypophosphatemia and hypomagnesemia can also be present in cows with hypocalcemia and can complicate response to treatment.

Calcium Dynamics of the Peri parturient Cow

During the last weeks of gestation, based on the requirements outlined in Chapter 7, a 650-kg dairy cow consuming 12 kg of DM needs to absorb about 24 g Ca each day for body maintenance (11 g) and for fetal development (13 g Ca/d). The average Holstein cow produces about 7 kg of first-milking colostrum (Mann et al„ 2016) with 2.1 g Ca/kg (see Table 12-1), representing 16 g Ca removed from the plasma. The Ca concentration in colostrum and transition milk exceeds 2 g/L for at least the first five milkings postpartum (Abd El-Fattah et al., 2012), which, combined with a first five milking yield of about 44 kg (Andrée O'Hara et al., 2019), represents a removal of about 88 g of Ca or 35 g/d with twice-daily milking. This equals a 4-fold increase in Ca requirements compared to the immediate prepartum cow. The exchangeable plasma pool of Ca in an early lactation cow is about 10 mg/kg BW (Ramberg et al., 1970). In a 650-kg BW cow, that pool must be turned over more than five times daily during the first 2 or 3 days of lactation to meet the increased demand created by the mammary gland. As a result, most cows will experience a decline in blood Ca at the onset of lactation, but most cows successfully activate Ca homeostatic mechanisms to return blood Ca concentration back to normal levels shortly after calving.

Ca homeostasis is mediated primarily by the parathyroid gland, which secretes parathyroid hormone (PTH) in response to any reduction in blood Ca concentration. The PTH stimulates release of Ca from bone stores, reduces the amount of Ca lost via urine, and activates the renal enzyme that produces the vitamin D hormone, 1,25-dihydroxy vitamin D. That hormone stimulates transcellular absorption of Ca across the intestinal epithelium to greatly increase uptake of dietary Ca (Goff, 2018). Serotonin is also involved with Ca metabolism perhaps via effects on PTH-related peptide (Hernandez et al., 2012). Infusing serotonin intravenously into prepartum cows increased concentrations of blood Ca postpartum and improved some other measures of Ca status (Weaver et al., 2016). A reduced ability of bone and kidney cells to respond to PTH stimulation has been implicated as the defect in Ca homeostasis that results in prolonged or severe hypocalcemia (Martig and Mayer, 1973). Several factors can interfere with Ca homeostasis, causing more drastic and longer-lasting declines in blood Ca, including age of the cow and breed (Lean et al., 2006; Roche and Berry, 2006; Chiwome et al., 2017).

Heifers rarely develop clinical milk fever, although 25 percent may experience subclinical hypocalcemia. The incidence of clinical and subclinical hypocalcemia increases with each subsequent lactation (Reinhardt et al., 2011; Venjakob et al., 2017). Ca homeostasis in heifers is more robust than in older cows because they are still growing and their bones contain more osteoclasts; hence, PTH only needs to activate the cells. This provides them a larger pool of exchangeable bone Ca on the first day of lactation (Ramberg, 1995). In older cows, the PTH must first induce osteoclast production and then activate the cells to resorb bone Ca. Also, as cows age, there is a reduction in the number of intestinal receptors for 1,25-dihydroxy vitamin D, which may translate into slower activation of intestinal Ca transcellular absorption (Horst et al., 1990).

The Channel Island breeds (Jersey, Guernsey) and, to a lesser extent, Swedish Red and White and Norwegian Red breeds have a higher incidence of milk fever than Holsteins (Lean et al., 2006; Chiwome et al., 2017). The reasons remain unclear, but Jerseys may have greater Ca stress because their colostrum has about 20 percent greater Ca concentration than that from Holsteins.

Diet Cation-Anion Difference and Acid-Base Status

Dietary cations, such as K^+, Na^+, Ca^{++}, and Mg^{++}, will raise blood pH when they are absorbed into the blood, and anions, such as chloride (CL), sulfate (SO_4^{2-}), and phosphate (PO_4^{3-}–), have the opposite effect. The difference in the number of milliequivalents of cations and anions absorbed from the diet helps determine blood pH (Stewart, 1983; Goff, 2018). Cows are typically in a state of compensated metabolic alkalosis as their diet consists of forages that are typically high in K. K is absorbed from the diet with nearly 100 percent efficiency, and because forage K often has an organic acid as the counterion, it is strongly alkalinizing. Grasses and legumes, especially those grown on soils where manure or potash has been applied, are usually major sources of dietary K (Pehrson et al., 1999). Na from compounds that lack an inorganic counterion (e.g., sodium bicarbonate) is also highly alkalinizing as it is absorbed with nearly 100 percent efficiency. Dietary concentrations of Ca and magnesium (Mg) can be high, but they are absorbed with much lower efficiency than K and Na (see Chapter 7) and therefore are less alkalinizing.

Adding CL and SO_4^{2-} without Na or K to the precalving diet can greatly reduce the degree of hypocalcemia at calving (Enderet al., 1971; Block, 1984; Oetzel et al., 1988). Cows in a state of compensated metabolic alkalosis do not respond to PTH stimulation as well as cows placed in a state of compensated metabolic acidosis (Goff and Horst, 1997b; Goff et al., 2014). Metabolic alkalosis impairs bone Ca resorption (Abu Damir et al., 1994; Block, 1994) and the ability of PTH to stimulate timely production of 1,25-dihydroxy vitamin D (Goff et al., 1991; Phillippo et al„ 1994).

Dietary CL is absorbed with nearly 100 percent efficiency. Sulfate anions can also acidify the blood but, because of lower absorption SO_4^{2-}, has just 60 percent of the acidifying activity of CL (Spears et al., 1985; Tucker et al., 1991; Goff et al., 2004). When sufficient anions are added to the precalving diet, they induce a compensated metabolic acidosis (Ender et al., 1971; Block, 1984), improving tissue sensitivity to PTH. This restores the competency of Ca homeostatic mechanisms and facilitates a rapid return to normocalcemia after the onset of lactation. Blood pH is difficult and expensive to measure, but urine pH generally reflects blood pH and can be measured on farm to determine the degree of compensated metabolic acidosis experienced by the cow. Diets that reduce urine pH values below 7.0 and usually closer to 6.0 generally improve Ca status (Charbonneau et al., 2006).

Several diet cation-anion difference (DCAD) equations (units of milliequivalents per kilogram of diet DM) have been developed (see Chapter 7). Although SO_4^{2-} is the acidifying anion, labs usually measure sulfur (S) so equations were developed using S. One meta-analysis found that the equation, DCAD = (Na + K) - (Cl+ 0.6 S), was best to predict urine pH and had the strongest association with milk fever incidence (Charbonneau et al., 2006). However, the equation, DCAD = (Na+ K) - (Cl + S), is probably the most widely used (DeGaris and Lean, 2008). Based on the meta-analysis of Charbonneau et al. (2006), a DCAD of about -200 mEq/kg (expressed as (Na+K) - (Cl + S)) is needed to achieve urinary pH of 6.5.

As DCAD decreases, the degree of hypocalcemia will also generally decrease (Moore et al., 2000; Charbonneau et al., 2006; Lean et al., 2006). If the addition of anions fails to acidify the blood enough to cause urine pH to be <7.0, there will be no improvement in periparturient Ca status (Moore et al., 2000; DeGaris and Lean, 2008; Leno et al., 2017; Goff and Koszewski, 2018). If the DCAD is too low, the cow can enter a state of uncompensated metabolic acidosis. As DCAD is reduced, the net amount of acid excreted into the urine will also increase until urine pH reaches about 6.3, but when urine pH is below 6.3, the net acid excretion in urine is no longer well correlated to DCAD (Constable et al., 2009). When urine pH falls below 6.3, the kidney begins to excrete NH_4^+ into the urine, slowing a further decline in urine and blood pH. The NH_4^+ arises from the combining of H^+ with ammonia that diffuses into the tubular fluid as urine pH is reduced below 5.9. This ability of the kidney to neutralize the excess H^+ allows the cow to remain in a state of compensated metabolic acidosis until urine pH reaches approximately 5.5 (Teloh et al., 2017). Urine pH below 5.3 is indicative of uncompensated metabolic acidosis (Berend, 2017), which will greatly reduce DMI (Goff, 2014). There is evidence that dry cows with urine pH of 7.3 will be less hypocalcemic than cows with urine pH of 7.9 but will exhibit more hypocalcemia than cows with urine pH of 6.0 (Moore et al., 2000). There is little practical difference in the degree of hypocalcemia experienced by cows with urine pH of 5.5 versus 6.7 (Charbonneau et al., 2006; Lean et al., 2006; Melendez and Poock, 2017).

The proper concentration of dietary Ca when cows are fed negative DCAD diets is not known. Successful anionic diets have contained between 0.65 and 1.7 percent Ca (Ender et al., 1971; Block, 1984; Gaynor et al., 1989; Joyce et al., 1997) . Once the dietary Ca requirement of the cow has been met, the addition of Ca to precalving diets has little effect on periparturient Ca status if blood has been acidified to the same extent. Dietary Ca, especially when added as calcium carbonate, has a mild alkalinizing effect that will necessitate addition of more anions to achieve the same degree of acidification, which may decrease DMI (Goff and Horst, 1997a; Goff and Koszewski, 2018). DMI is often reduced by low DCAD because of palatability (Oetzel et al., 1991) or by metabolic effects (Zimpel et al., 2018). Commercial anion supplements have been developed that are more palatable than traditional chloride or sulfate salts (Strydom et al., 2016).

Low-Calcium Diets to Prevent Hypocalcemia

This strategy involves limiting absorbed Ca so that the cow is in negative Ca balance for at least 7 to 14 days before calving. Negative Ca balance stimulates secretion of PTH within 3 to 4 days of the dietary Ca reduction, and PTH concentrations will remain elevated until after calving (Goings et al., 1974). Prolonged exposure to high concentrations of PTH overcomes any tissue resistance to PTH caused by metabolic alkalosis (Goff et al., 1986; Liesegang et al., 1998) . Ca conservation and Ca mobilization mechanisms, such as renal production of 1,25-dihydroxy vitamin D and bone Ca resorption, are activated prior to the onset of lactation (Boda and Cole, 1954; Goings et al., 1974; Green et al., 1981) so that homeostatic mechanisms are primed and ready to respond to the Ca demands of lactation.

Based on current Ca requirements and an availability coefficient of 0.44 for a high-forage diet (see Chapter 7), to meet the needs of a late-gestation 650-kg cow, the diet would have to contain about 52 g of Ca or 0.43 percent Ca (based on an intake of 12 kg). The studies in which milk fever was effectively prevented using the low dietary Ca approach had dietary Ca below 18 g/d (Boda and Cole, 1954; Goings et al., 1974; Green et al., 1981; Kichura et al., 1982) and likely supplied just 5 to 9 g of absorbable Ca each day. Based on feedstuffs typically available, feeding prepartum diets that are truly deficient in Ca is extremely difficult.

Substances that bind dietary Ca preventing absorption can cause a Ca deficiency and reduce periparturient hypocalcemia. Zeolite A, a sodium-aluminum silicate, binds Ca preventing absorption and can prevent hypocalcemia in cows fed diets with 0.6 to 0.7 percent Ca during the late dry period (Thilsing-Hansen et al., 2002; Pallesen et al., 2008; Kerwin et al., 2019). It also binds phosphate ($P0_4^{-3}$) (Pallesen et al., 2008), which may help prevent hypocalcemia (see below). Zeolite A also may bind Mg in the diet, resulting in lower plasma Mg concentrations (Thilsing-Hansen et al., 2002). However, when the prepartum diet contained more than 0.21 percent Mg, zeolite did not affect plasma Mg (Pallesen et al., 2008). Phytic acid in rice bran treated with formaldehyde is able to escape the rumen and bind Ca within the lumen of the small intestine, preventing it from being absorbed and improving periparturient Ca status (Martln-Tereso et al., 2016). However, average blood Ca concentrations the first 12 hours postcalving were still generally less than about 2.1 mmol/L.

Effect of Dietary Phosphorus on Hypocalcemia

Because it is an anion, absorbed dietary phosphate (PO_4^{-3}) will acidify the blood; however, excess dietary phosphorus (P) in the prepartum period increases the degree of hypocalcemia (Kichura et al., 1982; Barton et al., 1987). Serum $P0_4^{-3}$ generally increases as dietary P increases (Lopez et al., 2004), and this causes bone cells to secrete a PO_4-^{3} regulating hormone, fibroblast growth factor 23 (FGF23), to reduce blood $P0_4^{-3}$. The FGF23 circulates in the blood and binds to its receptor on kidney cells and inhibits renal synthesis of 1,25-dihydroxy vitamin D, which then reduces intestinal PO_4-^{3} absorption, causing blood PO_4^{-3} to decline. Unfortunately, lesser production of 1,25- dihydroxy vitamin D also reduces diet Ca absorption, impairing Ca homeostasis (Martin and Quarles, 2012). Restricting P intake to just meet requirement can aid Ca homeostasis. Cows fed an anionic precalving diet that was 0.21 percent P (approximately equal to requirement) had a lower incidence of hypocalcemia than cows fed a 0.44 percent P diet, and the low P diet maintained serum P concentrations that were within the normal range for cows (4 to 6 mg P/dL or 1.23 to 1.86 mmol/L) (Peterson et al., 2005). In a limited study, dairy cows fed P-deficient diets (i.e., plasma inorganic P concentrations were <1.0 mmol/L) the last 4 weeks of gestation had less clinical hypocalcemia than cows fed adequate P (Cohrs et al., 2018).

Dairy cows can develop a condition known as the hypophosphatémie downer cow. These cows exhibit a very low serum PO_4^{-3} concentration—often below 1.5 mg/dL (0.46 mmol/L). It is generally a complication associated with milk fever and is not a consequence of low dietary P concentrations. The mechanism is not well understood. It could involve reduced P intake because of low DMI that occurs during clinical hypocalcemia or perhaps sequestration of salivary phosphate in the rumen because of the reduced rumen motility that occurs during hypocalcemia. Secretion of PTH caused by low blood Ca increases loss of P via urine but can also increase P mobilization from bone and enhance gut absorption, so normally elevated PTH does not markedly reduce blood P concentrations. Steps taken to reduce milk fever incidence reduce incidence of the hypophosphatemic downer cow as well (Goff, 2014).

Dietary Magnesium and Hypocalcemia

Hypomagnesemia can contribute to hypocalcemia (Allen and Davies, 1981; Van de Braak et al., 1987). Blood Mg concentration is normally 1.9 to 2.4 mg/dL (0.8 to 1 mmol/L), but if blood Mg concentration falls below 1.25 mg/dL (0.5 mmol/L), the ability of the parathyroid gland to secrete PTH is compromised and blood Ca concentration rapidly decreases (Littledike and Goff, 1987). The effect of low Mg on PTH secretion is likely via Mg effects on guanosine diphosphate (GDP) dissociation from receptor proteins (Vetter and Lohse, 2002). This is most common in lactating dairy and beef cattle on pasture and is often referred to as lactation tetany or grass tetany. A less severe decline in blood Mg below 1.7 mg/dL (0.7 mmol/L) can alter the responsiveness of tissues to PTH (Contreras et al., 1982; Littledike and Goff, 1987; Rude et al., 2009). Cows fed adequate dietary Mg in the prefresh period will be slightly hypermagnesemic the day after parturition because of the actions of PTH on reabsorption of Mg from renal tubular fluid. Blood Mg concentration within 24 hours after calving that is equal to or less than 2.0 mg/dL (0.83 mmol/L) suggests inadequate dietary Mg (Goff, 2014). Based on a meta-analysis, an increase in dietary Mg concentration from 0.3 to 0.4 percent of DM, while maintaining DCAD and Ca constant, could result in an approximate 62 percent decrease in milk fever risk (Lean et al., 2006).

Vitamin D and Its Metabolites and Hypocalcemia

Meeting recommendations for dietary vitamin D (see Chapter 8) provides all of the substrate needed for adequate renal synthesis of 1,25-dihydroxy vitamin D. Feeding or injecting up to 10 million units of vitamin D between 4 and 14 days prior to calving can have a pharmacologic effect on Ca and P metabolism and prevent milk fever (Yamagishi et al., 2000). The bulk of the vitamin D is converted to 25-hydroxy vitamin D and other 24-hydroxylated vitamin D metabolites. The greatly elevated concentrations of 25-hydroxy vitamin D allow it to enter target cells. The 25-hydroxy vitamin D has much lower affinity for the vitamin D receptor than does 1,25-dihydroxy vitamin D; however, at greatly elevated concentrations, it will activate the receptor. The elevated 25-hydroxyvitamin D concentrations also displace 1,25-dihydroxy vitamin D from plasma vitamin D binding protein, raising the concentration of free 1,25- dihydroxy vitamin D (Jones, 2008). This will increase intestinal Ca absorption and can help prevent milk fever. Unfortunately, the dose of vitamin D that effectively prevents milk fever is very close to the toxicity level causing metastatic calcification of soft tissues. Lower doses may induce milk fever because the high levels of 25-hydroxyvitamin D and resulting hypercalcemia and hyperphosphatemia suppress renal synthesis of endogenous 1,25-dihydroxy vitamin D (Littledike and Horst, 1982). Several attempts have been made to feed or inject 25-hydroxyvitamin D prior to calving to prevent hypocalcemia, but they have not been consistently effective (Olson et al., 1973; Taylor et al., 2008; Weiss et al., 2015; Martinez et al., 2018; Rodney et al., 2018).

Experimental treatment with 1,25-dihydroxy vitamin D and its synthetic analogues can be more effective than using the less active vitamin D metabolites, but problems with timing of administration make these treatments impractical (Gast et al., 1977; Hove and Kristiansen, 1982). For effective prevention of milk fever, 1,25-dihydroxy vitamin D had to be administered between 7 and 3 days before calving. However, recently, a single dose of 1,25-dihydroxy vitamin D given within a few hours of calving improved periparturient Ca status when cows were fed an acidifying diet prior to calving (Viera-Neto, 2017). Exogenous 1,25-dihydroxy vitamin D also may have inhibited endogenous production of 1,25-dihydroxy vitamin D, so some cows developed hypocalcemia 5 to 12 days after calving (Horst et al., 2003). Continuous parenteral or oral administration of smaller doses of 1,25-dihydroxy vitamin D analogues prior to calving and for several days after calving can effectively prevent milk fever (Goff and Horst, 1990; Junichiro et al., 2015; Bachmann et al., 2017). A key to the success of these studies was that the daily hormone dose was reduced slowly after calving, allowing the cow to begin to make her own 1,25-dihydroxy vitamin D.

Displaced Abomasum

Displacement of the abomasum (DA) is a costly (Liang et al., 2017) multifactorial disorder and is diagnosed almost exclusively in adult dairy cattle but may occur in calves and young cattle (Zerbin et al., 2015; Biggs and Harvey, 2016; Caixeta et al., 2018). Average herd incidence rate is 2.2 percent (NAHMS-USDA, 2018), but individual herds can have rates as high as 20 percent (Doll et al., 2009). The transition period and several weeks subsequent to calving is the major risk period for development of displaced abomasum (Stengarde et al., 2010). About 80 percent of cases involve displacement to the left side of the cow and generally occur within the first 4 weeks following parturition (Radostits and Done, 2007). Only about 50 percent of the right DA occurs during this time (Zerbin et al., 2015). The causes of left and right DA are thought to be similar (Doll et al., 2009). Although the nature of relationships and associations is not well defined, twins, dystocia, milk fever, retained placenta, metritis, ketosis, and fatty liver are among the risk factors commonly identified for DA (Geishauser et al., 2000; Van Winden et al., 2003; LeBlanc et al., 2005).

Abomasal Physiology

In the nonpregnant cow, the abomasum occupies the ventral portion of the abdomen, very nearly on the midline, with the pylorus extending to the right side of the cow caudal to the omasum (Dyce et al., 1987; Radostits and Done, 2007). As pregnancy progresses, the growing uterus occupies an increasing volume of the abdominal cavity. The uterus begins to slide under the caudal aspect of the rumen, reducing rumen volume by one-third by the end of gestation. This forces the abomasum forward and slightly to the left side of the cow, although the pylorus continues to extend across the abdomen to the right side of the cow. After calving, the uterus retracts back toward the pelvic inlet, which, under normal conditions, allows the abomasum to return to its original position. In the case of a left DA, the pyloric end of the abomasum slides completely under the rumen to the left side of the cow (Van Winden et al., 2002). Three major factors thought to be responsible for DA are (1) the rumen fails to take up the void left by the retracting uterus, and if the rumen moves into its normal position on the left ventral floor of the abdomen, the abomasum is not able to slide under it; (2) the omentum attached to the abomasum is stretched, permitting movement of the abomasum to the left side; and (3) abomasal atony. Normally, gases produced in the abomasum (mostly carbon dioxide released when bicarbonate from the rumen meets the hydrochloric acid of the abomasum) are expelled back into the rumen as a result of abomasal contractions. These contractions are thought to be impaired in cows developing DA (Breukink and de Ruyter, 1976; Goff and Horst, 1997b; Doll et al., 2009). The exact causes of abomasal atony have not been fully established. Overconditioned cows at dry-off are at greater risk of DA likely because of low DMI around parturition (Cameron et al., 1998). In addition, excess loss of body condition from calving to 4 weeks postpartum is a risk factor for DA (Hoedemaker et al., 2009).

Nutrition and Abomasal Displacement

Reduced DMI before and after calving is likely a cause of DA, although direct evidence is limited (Shaver, 1997). Cows that eventually develop a DA have lower DMI a few days prior to clinical diagnosis (Van Winden et al., 2003). Poor feed bunk management (defined as <30 cm of bunk space per cow, feed refusal not removed daily, and empty feed bunks for a portion of the day) prepartum increased the risk of DA and also likely reduced DMI, but that was not measured (Cameron et al., 1998). Shaver (1997) outlined several factors that can increase the risk of DA and also likely limit or reduce DMI. These include limited feed availability, crowded pens and feed bunks, improperly mixed total mixed ration (TMR), sorting, and inclusion of unpalatable ingredients in the diet, among other factors.

Ca is needed for proper smooth muscle contractility and neuromuscular transmission (Caixeta et al., 2018); therefore, hypocalcemia is a risk factor for DA (Curtis et al., 1983; Rodriguez et al., 2017). Experimental induction of hypocalcemia reduced the rate of abomasal contractions, which may lead to atony and distension of the abomasum (Daniel, 1983). When plasma Ca levels were reduced from about 9.5 to 7.5 mg/dL, abomasal motility was reduced by 30 percent and strength of contractions was reduced by 35 percent, and when plasma Ca was reduced to 5 mg/dL, these responses were reduced to 70 percent and 50 percent, respectively (Daniel. 1983). Subclinical hypocalcemia increased the likelihood of DA 3.7 times (Rodriguez et al., 2017). The administration of oral $CaCl_2$ at calving to reduce subclinical hypocalcemia decreased the incidence of DA (Oetzel, 1996). However, the association between hypocalcemia and DA has not always been observed (LeBlanc et al., 2005; Chamberlin et al., 2013).

Increasing the proportion of grain in the diet fed to cows in late gestation and early lactation may increase the incidence of displaced abomasum (Coppock, 1974; Van Winden et al., 2004). This may be caused by increased volatile FAs within the abomasum, which can reduce abomasal contractility (Lester and Bolton, 1994). Elevated osmolality of rumen contents when high-grain diets are fed may contribute to paralysis and ultimately displacement of the abomasum (Van Winden et al., 2004). In addition, higher-grain diets may not supply adequate effective fiber needed to stimulate rumination activity, maintain the consistency and depth of the rumen mat, and illicit rumen contractions. Inadequate supply of effective fiber is likely a risk factor for DA. In an observational study, reduced rumination time both prior to and after calving was associated with greater incidence of DA (Stangaferro et al., 2016); however, controlled research with adequate statistical power evaluating the influence of effective fiber on incidence of DA is lacking. Direct research on the influence of TMR particle size, forage fiber, and fiber concentrations on rumen contractility and DA, especially during the transition period, is needed (Caixeta et al., 2018). For further information on effective fiber, see Chapter 5.

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